Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing
Department of CSE-Cyber Security

10211CC224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 24-04-2026

Employee Managment

SDG: SDG No – SDG Title(Example: SDG 4 –Employee Managment)

Presented By:

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Courses Handling Faculty:

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INTRODUCTION

1 An Employee Management System is used to store and manage employee data in an organization.

It helps maintain details like personal information, job roles, salary, and attendance.

2 The system replaces manual record-keeping with a digital and automated process.

3 It reduces errors and improves the accuracy of employee information.

4 It saves time by simplifying tasks like payroll, attendance tracking, and reporting.

5 The system provides quick and easy access to employee data when needed.

6 It improves overall efficiency and productivity of the organization.

7 It ensures better data security compared to traditional paper records.

8 It supports decision-making by providing organized and up-to-date information.

PROBLEM STATEMENT

1. Employee data is often managed manually, leading to inefficiency.
2. Manual record-keeping is time-consuming and difficult to maintain.
3. There is a high chance of errors in storing and updating employee information.
4. Lack of a centralized system makes data access slow and complicated.
5. Managing attendance, payroll, and employee details becomes challenging.
6. Data redundancy and duplication may occur in traditional systems.
7. There is a risk of data loss due to improper record handling.
8. Security of employee information is not well maintained in manual systems.
9. Overall organizational productivity is affected due to inefficient management processes.

OBJECTIVE(S)

1. To develop a centralized system for storing employee information.
2. To automate employee data management processes.
3. To reduce manual work and minimize human errors.
4. To efficiently manage attendance and payroll details.
5. To provide quick and easy access to employee records.
6. To ensure data accuracy and consistency.
7. To improve security of employee information.
8. To enhance overall organizational productivity.
9. To support better decision-making through organized data.

PROPOSED SYSTEM

1. The proposed system is a computerized application designed to manage employee data efficiently.
2. It provides a centralized database to store all employee-related information.
3. The system automates tasks such as attendance tracking, payroll, and record management.
4. It reduces manual errors and improves accuracy in data handling.
5. The system allows easy access and quick retrieval of employee information.
6. It includes secure login features to protect sensitive data.
7. The system ensures proper data backup to prevent data loss.
8. It improves overall efficiency and reduces time consumption in administrative tasks.
9. The proposed system enhances decision-making by providing organized and up-to-date reports.

PROPOSED SYSTEM ARCHITECTURE

1. The system follows a **three-tier architecture**: Presentation Layer, Application Layer, and Database Layer.
2. The **Presentation Layer (UI)** provides an interface for users (Admin/Employee) to interact with the system.
3. The **Application Layer (Business Logic)** processes user requests and performs operations like validation and calculations.
4. The **Database Layer** stores all employee-related data such as personal details, attendance, and salary.
5. The system uses a **client-server model**, where users access the application through a client interface.
6. A **secure login module** is included to authenticate users and control access.
7. The architecture supports **modular design**, including modules like Employee, Attendance, Payroll, and Reports.
8. Data flows from the user interface to the database through the application layer, ensuring proper validation.
9. The system ensures **data security, scalability, and reliability** through structured architecture design.

WORKFLOW

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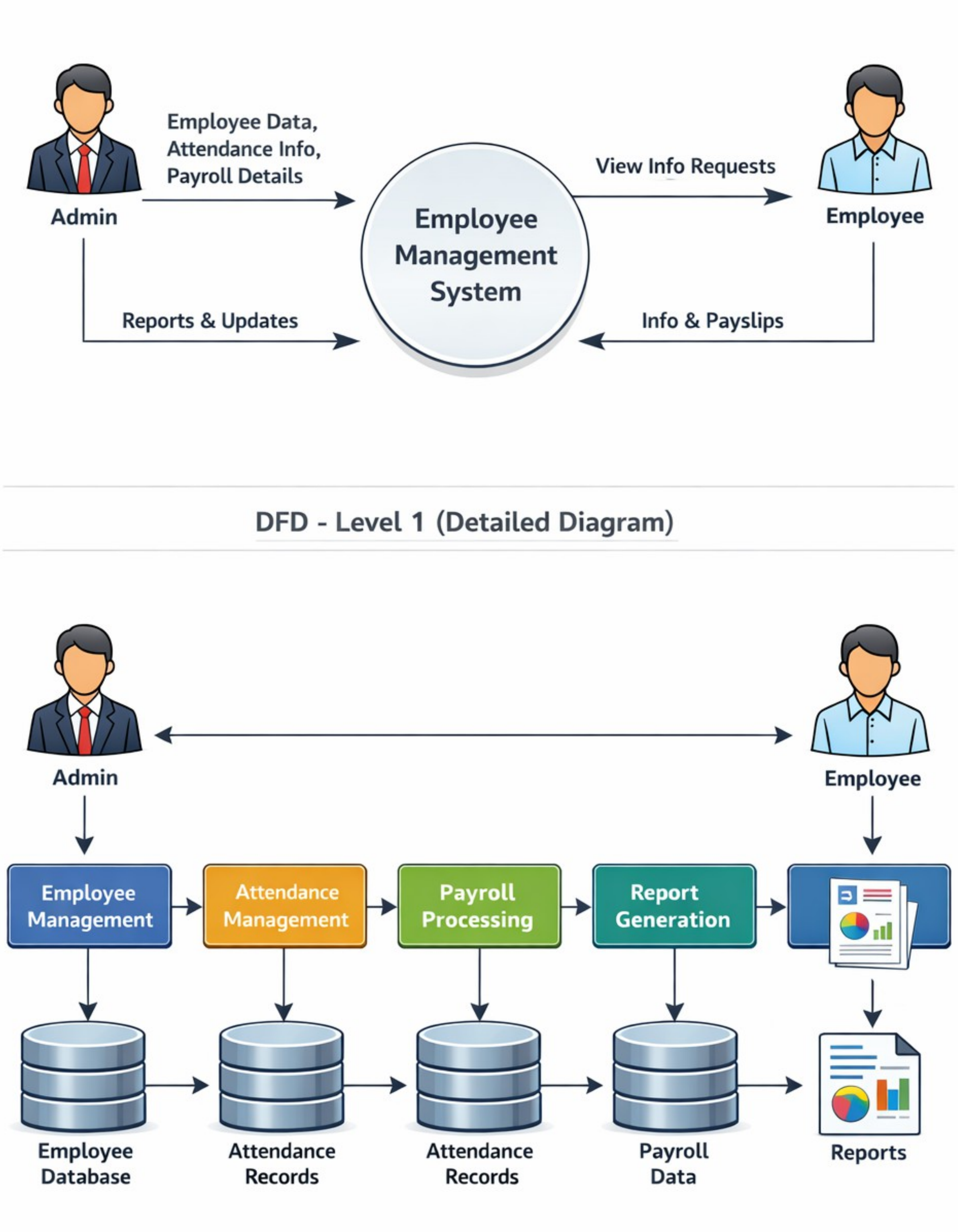

TECHNOLOGIES USED

1. **Java** – Used for developing the core application logic and backend processing.
2. **Spring Boot** – Framework used to build scalable and efficient backend services.
3. **Hibernate (ORM)** – Used for database interaction and object-relational mapping.
4. **MySQL** – Database used to store employee details, attendance, and payroll data.
5. **HTML** – Used to design the structure of web pages.
6. **CSS** – Used for styling and improving the user interface design.
7. **JavaScript** – Used to add interactivity and dynamic behavior to the application.
8. **JDBC** – Used for connecting Java applications with the database.
9. **Apache Tomcat** – Server used to deploy and run the web applicatio

MODULE DESCRIPTION

1. **Login Module** – Handles user authentication and provides secure access for Admin and Employees.
2. **Employee Management Module** – Used to add, update, delete, and view employee details.
3. **Attendance Management Module** – Records and tracks employee attendance on a daily basis.
4. **Payroll Module** – Calculates employee salary based on attendance, leaves, and other factors.
5. **Leave Management Module** – Allows employees to apply for leave and admin to approve/reject requests.
6. **Department Management Module** – Manages different departments and assigns employees accordingly.
7. **Report Generation Module** – Generates reports such as employee details, attendance, and salary slips.
8. **Database Management Module** – Handles storage, retrieval, and maintenance of all system data.
9. **Security Module** – Ensures data protection through authentication, authorization, and data backup.

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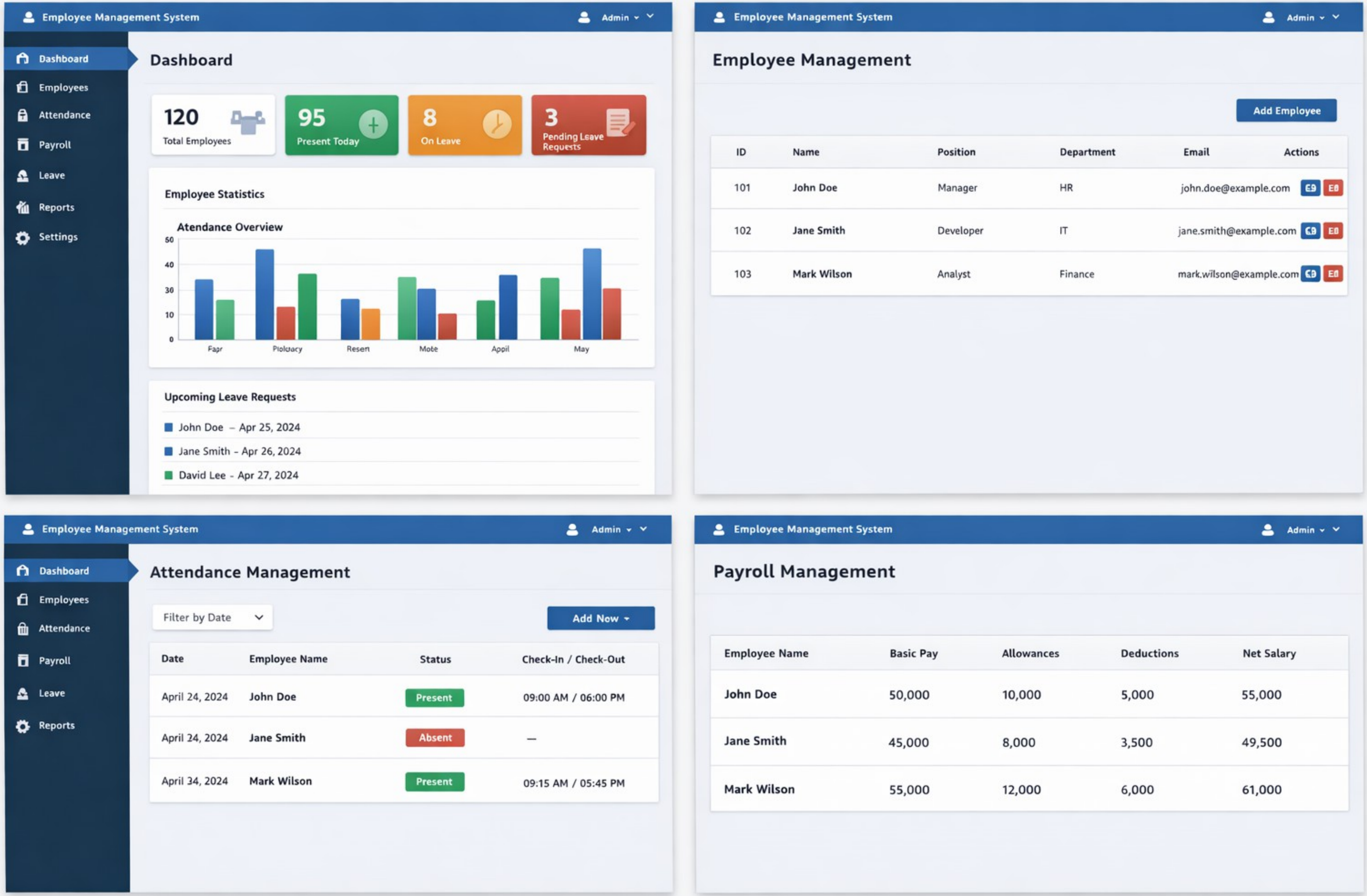
Employee Management System
Admin
Dashboard
Employees
Attendance
Payroll
Leave
Reports
Settings
Dashboard
120
Total Employees
95
Present Today
8
On Leave
3
Pending Leave Requests
Employee Statistics
Atendance Overview
50
40
30
10
0
April
May
Upcoming Leave Requests
John Doe – Apr 25, 2024
Jane Smith – Apr 26, 2024
David Lee – Apr 27, 2024
Employee Management System
Admin
Employee Management
Add Employee
ID
Name
Position
Department
Email
Actions
101
John Doe
Manager
HR
john.doe@example.com
102
Jane Smith
Developer
IT
jane.smith@example.com
103
Mark Wilson
Analyst
Finance
mark.wilson@example.com
Employee Management System
Admin
Dashboard
Employees
Attendance
Payroll
Leave
Reports
Attendance Management
Filter by Date
Add Now
Date
Employee Name
Status
Check-In / Check-Out
April 24, 2024
John Doe
Present
09:00 AM / 06:00 PM
April 24, 2024
Jane Smith
Absent
—
April 34, 2024
Mark Wilson
Present
09:15 AM / 05:45 PM
Employee Management System
Admin
Payroll Management
Employee Name
Basic Pay
Allowances
Deductions
Net Salary
John Doe
50,000
10,000
5,000
55,000
Jane Smith
45,000
8,000
3,500
49,500
Mark Wilson
55,000
12,000
6,000
61,000

ADVANTAGES & LIMITATIONS

1. Reduces manual work, but requires initial setup cost.
2. Improves data accuracy, but depends on proper system design.
3. Saves time in managing records, but needs regular maintenance.
4. Provides quick access to information, but requires technical knowledge.
5. Enhances data security, but may face security risks if not handled properly.
6. Automates attendance and payroll, but depends on system reliability.
7. Improves organizational efficiency, but requires user training.
8. Generates reports easily, but customization may be limited.
9. Supports decision-making, but depends on internet/system availability

Thank you